

Real Time Network Traffic Analysis using Artificial Intelligence, Machine Learning and Deep Learning: A Review of Methods, Tools and Applications

Aschin Dhakad

Electronics and Telecommunication
Engineering, RV College of
Engineering®, Bengaluru, India
aschindhakad.et20@rvce.edu.in

Shruti Singh

Electronics and Telecommunication
Engineering, RV College of
Engineering®, Bengaluru, India

Mohana

Computer Science & Engineering
(Cyber Security), RV College of
Engineering®, Bengaluru, India

Minal Moharir

Computer Science & Engineering
(Cyber Security), RV College of
Engineering®, Bengaluru, India

Ashok Kumar A R

Computer Science & Engineering
(Cyber Security), RV College of
Engineering®, Bengaluru, India

Abstract- In order to spot potential security threats or performance problems, Network Traffic Analysis (NTA) involves monitoring and analyzing network traffic. However, Machine Learning (ML) methods are frequently used to automate NTA. Network traffic classification, anomaly detection, and malicious activity detection can all be done using ML techniques. In order to enhance network performance, they can also be utilized to forecast future traffic patterns. ML algorithms come in a variety of forms and can be applied to NTA. Support vector machines (SVM), decision trees, and random forests are the most used methods. Depending on the particular application, an algorithm will be chosen. SVMs, for instance, are frequently used for classification tasks, whereas decision trees are frequently utilized for anomaly detection jobs. Network performance and security can be enhanced using NTA employing ML. It can aid in the detection of possible risks, the prevention of data breaches, and the enhancement of network performance. In the proposed work real time NTA using ML and DL algorithms discussed with tools and applications. Random forest algorithm is implemented and obtained an accuracy of 99.31%. Benefits of applying ML to NTA includes increased accuracy when it comes to spotting dangers and anomalies, ML algorithms have the potential to be more precise than conventional rule-based techniques. Less false positives, ML algorithms can be customized to produce fewer false positives, which can save time and money. Enhanced scalability, ML algorithms can be scaled to manage high levels of network traffic.

Keywords — *You Only Look Once (YOLO), Convolutional Neural Network (CNN), Artificial Intelligence(AI), Machine learning (ML), Deep Learning(DL)*

I. INTRODUCTION

The process of gathering, examining, and interpreting network traffic data in order to spot anomalies, performance problems, and malicious behavior is known as network traffic analysis. Conventional techniques for analyzing network traffic frequently rely on signature-based detection, which searches for patterns of malicious traffic that have been identified. This strategy may not work against emerging or novel dangers. Deep learning and

artificial intelligence (AI) provide up new avenues for real-time network traffic analysis. Without the requirement for pre-defined signatures, harmful traffic patterns can be recognized by AI-powered traffic analysis tools. They are hence more potent against emerging and changing threats. real-time NTA is a difficult undertaking reasons are first; network traffic is frequently both highly fast and big volume. Because of this, gathering and analyzing all of the traffic in real time is challenging. Secondly, a lot of network traffic is encrypted. This can make it challenging to detect malicious activity since it makes it harder to view the substance of the traffic. Third, traffic on networks is ever-changing. This implies that in order to stay abreast of the most recent dangers, the models used for traffic analysis must be updated on a regular basis. Real-time NTA exhibits a number of difficulties that can be addressed by AI and deep learning. First, without the need for pre-defined signatures, AI-powered traffic analysis systems can utilize machine learning to learn to recognize harmful traffic patterns. They are hence more potent against emerging and changing threats. Second, deep learning can be used by AI-powered traffic analysis systems to decipher encrypted traffic. DL models have the capacity to recognize encrypted traffic patterns linked to harmful behavior. Thirdly, compared to conventional traffic analysis tools, AI-powered solutions can receive updates more often. Due to the ease and speed with which AI models can be trained on new data sets.

II. LITERATURE SURVEY AND PROBLEM ANALYSIS

Neeraj Namdev *et al* [1] presented the use of the internet is playing a huge role as technology and communication systems evolve. As a result, data and traffic through the internet rise exponentially. A particularly common method for fighting the information detection system is the categorization of internet traffic. Nour Alqudah *et al* [2], presented that traffic analysis serves a variety of functions, including assessing the efficiency and security of network administration and operations. It is

believed that NTA is essential for enhancing the security and functionality of networks. New approaches are needed to identify intrusions, categorize Internet traffic, and analyze virus behavior as a result of growing network traffic and the development of artificial intelligence. A. Jamuna *et al* [3], presented the traffic classification based on applications creation is a key component of network security and management. Traditional methods include payload and port-based deep examination approaches. Several privacy concerns, dynamic ports, and encrypted programs make the conventional procedures ineffective in the current network context. T. Bujlow *et al* [4] presented that monitoring the network performance in high-speed internet infrastructure is a difficult undertaking because the standards for the specified quality level depend on the type of service. Understanding the different sorts of applications constituting the present network traffic is consequently necessary for backbone QoS monitoring and analysis in multihop networks. It was suggested to use the C5.0 ML Algorithm to improve traffic classification in order to address the shortcomings of the current approaches. Amin Shahraki *et al* [5] presented that for a variety of objectives, including network resource management and cyber-security analysis, this data analysis is crucial. Analytical techniques that can process network data online based on the receipt of new data. Data analytics are predicted to be supported by online ML (OL) approaches. A. Priya *et al* [6] presented that numerous additional techniques were suggested to address the current technical problems. The primary objective of network traffic classification is network performance improvement. The suggested framework states that this work uses unsupervised ML to identify the user's browser and application from real-time network information. V. A. Muliukha *et al* [7] presented that the application of map-reduce technology to NTA. It describes how to determine when a frame in a PCAP file begins. The procedure for producing training samples for various traffic types is explained. There is a description of the programme "Tractor" used in the experiments as well as the findings of experimental investigations on the analysis of encrypted traffic using the random forest method and the naïve Bayesian classifier. E. Nazarenko *et al* [8] presented that the theories underlying the operation of the traditional ML algorithms and the knowledge of their parameters were regarded as a component of their theoretical makeup. The created network traffic categorization and identification system makes use of the k-NN, Naive Bayes, and SVM ML algorithms. The theoretical underpinnings of traditional ML algorithms—their operating principles and parameter knowledge—were considered. E. Osa *et al* [9] presented that Network monitoring and unauthorized access or malicious traffic over secured networks are detected using intrusion detection systems. The comparative examination of a few ML techniques presented. ML methods in identifying anomalies in network data is compared in this research. Decision Tree was judged to provide the overall best outcome. L. Trajković *et al* [10] presented that Traffic traces gathered from operational communication that characterizes quantify traffic loads, examine user behavior patterns, model network traffic, and forecast network traffic in the forthcoming. To characterize and model network traffic, examine Internet topologies, and categorize network anomalies, use traffic traces gathered from various deployed networks and the Internet. Yang *et al* [11] presented that the foundation for network traffic monitoring, data analysis, and user service quality improvement is accurate

network traffic identification. DPI technology can also identify specific application traffic, which increases the accuracy of identification. The use of the ML method based on statistical flow characteristics helps identify network flows with encryption and unknown features, making up for the shortcomings of DPI technology. Y. Xue *et al* [12] presented that prior to discussing some suggestions that can improve traffic performance, first try to analyze the present traffic classification methodologies in this research, concentrating on their issues and challenges. A. Boukhalfa *et al* [13] presented that the detection of threats is complicated by the rising volume and diversity of data in these interactions. In order to identify new invisible dangers, this study employs a novel approach to NTA that leverages big data frameworks to gather substantial volumes of network traffic data and Deep Learning (DL) algorithms to analyze it. K. Limthong *et al* [14] conducted studies to look at the relationship between interval based characteristics of network traffic and various sorts of network abnormalities using two well-known ML techniques, naïve Bayes and k-nearest neighbor. two ML algorithms, five different kinds of test bed anomalies, and nine interval-based features of network traffic to analyze and assess each feature's correctness. Although only from the naïve Bayes and k- nearest neighbor algorithms, the preliminary results showed the more useful features for each of the anomaly types. M. Shafiq *et al* [15] presented that Internet traffic can be categorized using a variety of traditional techniques, including ML, payload analysis, and port-based classification. ML (ML) approach is currently the most often utilized method, which has received very accurate results and is used by many researchers. The limitations of the several network traffic classification techniques (port-based, payload-based, and ML-based) are presented. The network traffic classification model is then organized from traffic capture to final output. Used Wire Shark tool to record five WWW, DNS, FTP, P3P, and TELNET applications traffic for a length of one minute and the Netmate programme to extract 23 attributes for comparative study of four algorithms. M. Ramires *et al* [16] presented that the goal of traffic classification methods is to automatically analyze and classify traffic moving over a network based on its unique properties and intrinsic nature. Taking advantage of factors like port numbers and payload analysis for categorization, However, due to the Internet's rapid growth and evolution, such techniques have proven ineffective. a comparative analysis of ML approaches for categorization of network traffic. D. Szostak *et al* [17] presented that to provide services to users, network operators must prioritize efficient resource allocation. Its optimization provides for cost savings, required service quality, and anomalies in data flow detection. Linear Discriminant Analysis (LDA) classifier-based ML (ML) method for estimating short-term traffic numbers was offered. The traffic prediction problem is therefore formulated as a classification task. M. Soykan *et al* [18] presented that A product that enables anonymous Internet communication without disclosing users'; identity is the Tor project. first conducted data analysis and learned more about the data set. categorical values were allocated to numerical values. Iraj Lohrasbinasab *et al* [19] presented to address problems, network performance is tracked using Network Traffic Monitoring and Analysis (NTMA), a collection of approaches. Forecasting network load and its future behavior is the main focus of the significant NTMA 'Network Traffic Prediction' (NTP). Typically, one of two methods—statistical or ML—can be used to deploy NTP techniques.

Kuldeep & Agrawal *et al* [20] presented that Traditional IP traffic classification techniques like port number and payload-based direct packet inspection techniques are no longer widely used due to the use of dynamic port numbers in packet headers rather than well-known port numbers and various cryptographic techniques that prevent inspection of packet payloads. A contemporary trend is to categorize IP traffic using ML (ML) techniques. A packet capture used to build a real-time internet traffic dataset during a 2 second packet collecting period.

III. AI AND DEEP LEARNING METHODS IN NETWORK TRAFFIC ANALYSIS

A. Network Traffic Analysis - To detect and counteract security threats, improve network performance, and troubleshoot network issues, NTA involves gathering, monitoring, and analyzing data on network traffic. The application of AI and DL in NTA is growing as a means of automating processes, enhancing accuracy, and generating additional insights. Network traffic, including web traffic, email traffic, and file transfer traffic, can be categorized using AI. Possible security risks like malware or distributed denial-of-service (DDoS) attacks can be identified. The performance and capacity planning of network may be improved as a result by figuring out the source of the issue, AI can be utilized to diagnose network issues. Examples of specific applications of AI and DL in NTA include: Stealthwatch is an NTA solution that Cisco has created that uses AI. Stealthwatch employs ML to recognize and categorize network traffic, find abnormalities and traffic patterns. A NTA solution powered by AI named Wildfire was created by Palo Alto Networks. WildFire employs ML to recognize and categorize malware, recognize malicious traffic, and stop assaults. Contrail Analytics is an NTA solution developed using AI. ML is used by Contrail Analytics to recognize and categorize network traffic, find abnormalities, and improve network performance.

B. Classification methods

Supervised Learning - For NTA utilizing ML, supervised categorization techniques are frequently utilized. The model gains knowledge from labeled data to forecast instances of fresh, unforeseen network traffic. The following are a few well-liked supervised classification techniques for NTA.

Decision Trees - For categorization jobs, decision trees offer a straightforward but efficient approach. To make decisions, they divided the data into subgroups based on several attributes and built a tree like structure. The edges of the tree represent decisions based on the features represented by the nodes, which make up the tree. Random Forest: An ensemble method that mixes various decision trees is known as a random forest. To produce more precise predictions, it builds multiple decision trees during training and aggregates their results. Random forests increase the overall robustness and less overfitting.

SVM - SVM is an effective binary classification system that handles both linear and non-linear data with ease. It seeks to identify the hyperplane in the feature space that best divides the classes. One-vs-all or one-vs-one approaches, for example, can be used to extend SVM to handle multi-class classification. It uses a logistic regression model to estimate the likelihood of a particular class occurring depending on features k-Nearest Neighbors (k-NN) is a straightforward and understandable classification algorithm. When a new instance is encountered, the system examines the 'k' nearest labeled examples and assigns the class that shows up the most frequently among the 'k'

neighbors. Naive Bayes is a probabilistic classification technique built on the Bayes theorem. NTA can be done using neural networks, such as convolutional neural networks (CNNs). Although they need a lot of labeled data and computer power to train for complex data patterns. *Gradient Boosting* - Gradient boosting is another ensemble technique that sequentially constructs a number of weak learners (typically decision trees), each of which attempts to correct the mistakes of the previous ones. It is renowned for having a high degree of predictability. These well-liked gradient boosting algorithm implementations, XGBoost and LightGBM, have been scaled up and improved for performance. Since they are so effective, they are frequently used in many classification tasks. AdaBoost is an ensemble learning technique that concentrates on hard-to-classify instances. In the subsequent rounds, the weights are iteratively adjusted to give more weight to instances that were incorrectly classified. It assigns weights to the training instances. *Unsupervised Learning* - Unsupervised classification, also known as clustering, is a ML approach where the goal is to group similar instances together without any predefined labels. In the context of NTA, unsupervised classification methods aim to discover underlying patterns and structures in the data without using labeled examples. common unsupervised classification methods used by NTA.

Hierarchical Clustering: Hierarchical clustering creates a tree-like structure of nested clusters, also known as a dendrogram. It can be agglomerative (bottom-up) or divisive (top-down). At each step, the algorithm merges or splits clusters based on their similarity.

DBSCAN : density- based clustering algorithm that groups point together based on their density in the feature space. It can automatically identify outliers as noise points.

Gaussian Mixture Models (GMM): It estimates the parameters of distributions and assigns each point to the most probable cluster.

Self-Organizing Maps (SOM): Type of ANN that projects high-dimensional data onto a lower-dimensional grid. It organizes the data in a topological manner, grouping similar instances closer to each other on the grid.

Agglomerative Information Bottleneck (AIB): AIB is a clustering algorithm that finds the optimal trade-off between clustering and preserving information, useful for analyzing high-dimensional data like network traffic.

OPTICS (Ordering Points to Identify the Clustering Structure): OPTICS is an extension of DBSCAN that creates a reachability plot, which allows for flexible clustering based on different density levels.

BIRCH (Balanced Iterative Reducing and Clustering using Hierarchies): BIRCH is a hierarchical clustering method that constructs a tree-like structure to efficiently cluster large datasets. Unsupervised classification methods can be valuable for NTA, especially when there is little or no labeled data available or when exploring the data for potential patterns and anomalies. Additionally, domain expertise is often required to interpret the clustering results effectively and derive meaningful insights from the discovered patterns.

C. Online learning and offline learning

A ML model is trained online, with incremental updates made as new data comes in. As a result, the model is always evolving and modifying to account for new data. When doing real-time data streaming activities like fraud detection or spam filtering, online learning is frequently used. One way to train a ML model is through offline learning, which involves training the model on

static data. As a result, the model does not change as new data becomes available. In tasks like image classification or natural language processing, where the data is not streaming in real time, offline learning is frequently used.

IV. NETWORK TRAFFIC ANALYSIS TOOLS

NTA tools are software applications or platforms designed to monitor, capture, analyze, and visualize network traffic. some popular NTA tools are.

Wireshark: Wireshark is a widely used open-source packet analyzer that allows users to capture and inspect network packets in real-time.

tcpdump: tcpdump is a command-line packet capture tool available on Unix-like operating systems. It can capture and display network traffic in real-time or save it to a file for later analysis.

Ethereal: Ethereal, now known as Wireshark, is a packet analyzer that runs on multiple platforms and supports a broad range of protocols.

tshark: tshark is the command-line version of Wireshark, allowing users to perform packet analysis without a graphical interface.

NetFlow Analyzer: NetFlow Analyzer is a tool that collects and analyzes NetFlow data from network devices to provide insights into network traffic patterns and bandwidth usage.

PRTG Network Monitor: It monitors bandwidth usage, traffic patterns, and provides detailed reports.

Ntopng: Ntopng is a high-performance NTA tool that provides real-time and historical traffic data, including top talkers, protocols, and applications.

SolarWinds Network Performance Monitor: This tool offers NTA, monitoring, and alerting features to help administrators identify and resolve network issues.

Nagios: Nagios is a popular open-source network monitoring and alerting system that can be extended with plugins to perform traffic analysis.

Capsa Network Analyzer: Capsa is a user-friendly network analyzer that offers real-time and historical NTA, protocol analysis, and bandwidth monitoring.

Splunk: Splunk is a versatile platform used for log aggregation, analysis, and visualization, making it suitable for NTA when combined with appropriate plugins.

Observer (formerly Network Instruments Observer): Observer is a comprehensive network monitoring and analysis platform that provides in-depth traffic analysis capabilities.

FlowTraq: FlowTraq is a network security and traffic analysis tool that focuses on NetFlow and IPFIX data.

Graylog: Graylog is a log management and analysis platform that can be utilized for NTA when network logs are available. These tools offer various features and capabilities, so the choice of the right tool depends on the specific requirements and objectives of the network analysis tasks. Some tools are more suitable for real-time monitoring and alerting, while others focus on in-depth protocol analysis and historical data examination.

V. POPULAR ML AND DL ALGORITHMS AND FRAMEWORKS FOR NTA

A. ML algorithms

SVMs: SVMs are used for NTA because they learn complex relationships between the features of network traffic.

Decision trees: Decision trees are used for NTA because they are easy to interpret and to identify important features of network traffic.

B. DL algorithms

CNNs: CNNs are a type of DL algorithm that is well-suited for tasks that involve processing images or sequences of data. CNNs have been used for NTA to identify malicious traffic, such as malware or botnets.

Long short-term memory networks (LSTMs): LSTMs are a type of DL algorithm that is well-suited for tasks that involve processing sequential data. LSTMs have been used for NTA to identify patterns in network traffic that are indicative of malicious activity.

Generative adversarial networks (GANs): GANs are a type of DL algorithm used to generate synthetic data that is similar to real data. GANs have been used for NTA to generate synthetic network traffic used to train ML models.

C. Frameworks

TensorFlow: TensorFlow is a popular open-source framework for ML and DL. TensorFlow is well-suited for NTA because it is able to process large amounts of data quickly and efficiently.

PyTorch: PyTorch is another popular open-source framework for ML and DL. PyTorch is similar to TensorFlow, but it is more flexible and easier to use.

Scikit-learn: Scikit-learn is a popular Python library for ML. Scikit-learn includes a number of algorithms used for NTA, such as SVMs, decision trees, and random forests. The specific algorithms and frameworks that are used will depend on the specific task and application.

VI. IMPLEMENTATION OF REAL TIME NETWORK TRAFFIC ANALYSIS

Real-time NTA refers to the process of monitoring and analyzing network traffic as it occurs, providing immediate insights into the current state of the network. This type of analysis is crucial for detecting and responding to network anomalies, security threats, performance issues, and other events that require timely action. Real-time NTA typically involves the following steps:

Packet Capture: To perform real-time analysis, network packets need to be captured as they traverse the network. This can be achieved using packet capture tools like Wireshark or tcpdump, or by leveraging network devices that support packet capture functionalities.

Packet Filtering and Processing: As packets are captured, they need to be filtered and processed to extract relevant information and features. This step may involve filtering out irrelevant traffic, dissecting protocol headers, extracting payload data, and generating flow data for efficient analysis.

Feature Extraction: Important features are extracted from the network traffic data. These features may include source and destination IP addresses, ports, protocols, packet sizes, payload

types, packet timestamps, and more. These features are crucial for building models and making real-time decisions.

Real-Time Analysis: These analyses can include identifying abnormal patterns, detecting security threats like intrusion attempts, predicting network performance issues, or classifying traffic based on application or behavior.

Alerting and Visualization: If the analysis reveals any significant events or anomalies, real-time alerts are triggered to notify network administrators or security teams. Visualization tools are also used to provide real-time graphical representations of network traffic, making it easier to interpret and respond to the data.

Continuous Monitoring: Real-time NTA is an ongoing process that requires continuous monitoring of the network. The captured data is constantly updated, and the analysis is performed in real-time on the latest information.

Response and Mitigation: Based on the analysis and alerts, network administrators can take immediate action to mitigate threats, resolve performance issues, or investigate suspicious activities in real-time.

Real-time NTA is essential for ensuring network security, optimizing network performance, and maintaining the overall health of the network. It allows organizations to respond quickly to incidents and make informed decisions based on up-to-date network information.

Software Requirements - Python 3.11.4 was used as the high-level language to create the system's software. The Jupyter notebook environment was used for the development process. The model was developed using the Python language's pandas, scikit-learn, seaborn, matplotlib, and Numpy libraries.

- tcp/udp/icmp length- length after fragmentation
- protocol- if the protocol is used or not.
- label-if the label is used or not (virtual circuit approach or datagram approach)

VII. RESULTS AND ANALYSIS

Dataset specifications and features - The properties of network traffic utilized to distinguish and assess network activities are referred to as network traffic features. The following uses for these features are just a few examples: Security of the network: Features of network traffic can be exploited to spot malicious traffic, such as malware or botnets. Network troubleshooting: Issues with the network, like packet loss or latency, can be resolved by using the features of the network traffic. The network traffic features that are used are.

- SERVICE-the service value for its packets to a high value to ensure that the packets are routed quickly and reliably higher the no more the reliability.
- Sport- senders port address at application layer
- Dport- destination port address at application layer
- ttl-time to live
- ip_length- original length of the ipv4 packet
- ip_checksum- checksum value for error detection
- ip_id-IP-used to assist packet segmentation and reassembly.
- ip_offset- data in the prior bits divided by 8.

Random forest model is implemented. And obtained an accuracy of 99.31%.

```
training_predicted = rf.predict(train_features)
testing_predicted = rf.predict(test_features)

training_cm = confusion_matrix(train_labels, training_predicted.round())
testing_cm = confusion_matrix(test_labels, testing_predicted.round())
```

Fig.1 Code snippet for confusion matrix

```
rf = RandomForestClassifier(n_estimators = 100, max_depth = 4, criterion = "gini")
rf.fit(train_features, train_labels)

train_predict=rf.predict(train_features)
test_predict=rf.predict(test_features)
```

Fig.2 Code snippet for training and testing of random forest model.

```
scaler = MinMaxScaler()
features_train = scaler.fit_transform(train_features)
features_test = scaler.transform(test_features)
```

Fig.3.Code snippet for data normalization using scalar

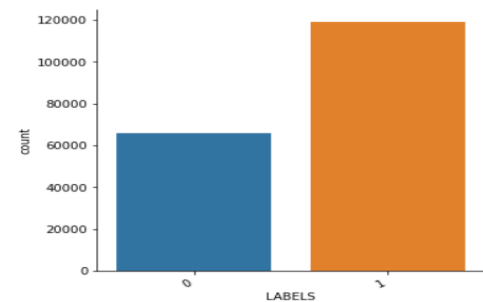


Fig.4.Data split about features

Figures 1 to 3 show the code snippet of training and testing of random forest model to generate confusion matrix. Figure 4 shows the details of data split features. It is a useful tool for understanding how well the algorithm is able to distinguish between different classes. Table 1 shows the confusion matrix.

TABLE 1. CONFUSION MATRIX OF TRAINING AND TESTING

52048	725	13009	184
315	94908	70	23737

Training accuracy = 99.29 %

Testing accuracy = 99.31%

The accuracy of a model, which considers both precision and recall. It is derived as the harmonic mean of recall and precision, giving both measures equal weight. In a multi-class classification task, averaging measurements over several classes is done using the macro average technique. It is an approach to combine the outcomes of a classification model across various classes without assigning any one class more importance. An average that considers the relative significance of various values in a data collection is called a weighted average. Each value is multiplied by a weight in a weighted average, and the total of the weighted values is then divided by the total of the unweighted values.

VIII. DESIGN AND IMPLEMENTATION CHALLENGES

Data volume and complexity-NTA typically involves collecting and analyzing large amounts of network traffic data. This data can be very complex, as it can include a variety of different protocols, packet types, and encryption levels.

False positives and negatives-NTA systems can generate false positives and negatives. False positives occur when the system incorrectly identifies benign traffic as malicious. False negatives occur when the system incorrectly identifies malicious traffic as benign.

Scalability-NTA systems scale to handle large amounts of network traffic. This can be a challenge, as NTA systems typically need to be able to process data in real time.

Cost - NTA systems can be expensive to implement and maintain and collect and analyze large amounts of network traffic data.

A. Applications

Network security: NTA used to mitigate security threats, such as malware, botnets, and denial-of-service attacks.

Network performance: NTA used for performance bottlenecks and optimize network performance.

Network troubleshooting: NTA used to troubleshoot network problems, such as packet loss or latency.

Compliance: NTA used to help organizations comply with regulations, such as those related to data privacy and security.

Forensics: To investigate security incidents and to gather evidence for legal proceedings.

Business intelligence: NTA used to gather insights into user behavior and to improve business decision-making.

B. Future research directions

NTA is a rapidly evolving area, and there are a number of open issues and future research directions. Some of these issues include:

Heterogeneity of network traffic - number of devices and applications connected to the internet increasing, the heterogeneity of network traffic is increasing. This makes it more difficult to develop NTA systems that can accurately identify and classify all types of network traffic.

Dynamic nature of network traffic - Network traffic is constantly changing, as new applications and protocols are developed and as users' behavior changes. This dynamic nature makes it difficult to design NTA systems that can keep up with the latest changes in network traffic.

Privacy concerns - NTA systems collect and analyze large amounts of network traffic data, which can raise privacy concerns for users.

Cost - NTA systems can be expensive to implement and maintain.

IX. CONCLUSION

NTA using ML is a powerful and essential approach for gaining valuable insights, enhancing security, optimizing performance, and automating network management. By harnessing the capabilities of ML algorithms, organizations can effectively process and interpret the vast amount of network traffic data generated every day. This data-driven approach enables us to make informed decisions, respond rapidly to security threats, and improve the overall efficiency of their networks. real time NTA implemented using random forest algorithm and obtained an accuracy of 99.31%. Some key takeaways from NTA using ML are Enhanced Security-ML models can detect and prevent various security threats, including malware, intrusions, and denial-of-service attacks, by analyzing network traffic patterns and identifying suspicious activities in real-time. Anomaly Detection-ML algorithms can identify abnormal network behavior and performance deviations, enabling prompt action and

minimizing downtime caused by network issues. Optimized Performance-through predictive analysis and traffic engineering, ML can help optimize network performance, capacity planning, and QoS management, ensuring a seamless user experience. Automated Network Management- ML models integrated into network management systems can automate decision-making processes, load balancing, and resource allocation, reducing manual intervention and improving network efficiency. Insights into User Behavior-ML analysis provides valuable insights into user behavior, allowing organizations to detect potential insider threats or suspicious activities. Efficient Traffic Classification-ML can accurately classify network traffic into different application types, aiding in bandwidth management and prioritizing critical applications. Network Forensics and Predictive Maintenance-ML assists in network forensics, enabling the investigation of past incidents and predicts equipment failures to support proactive maintenance.

Despite the numerous benefits, NTA using ML also poses some challenges. Properly handling and preprocessing large-scale network data, selecting appropriate features, dealing with imbalanced datasets, and avoiding overfitting are some of the challenges that need to be addressed. To leverage the full potential of NTA using ML, continuous research and development in the field of artificial intelligence, data science, and cybersecurity are vital. Additionally, collaboration between domain experts and data scientists is essential to design effective models tailored to specific network environments and use cases. As technology advances and new ML techniques emerge, NTA will continue to evolve, playing a pivotal role in securing networks, improving performance, and enabling more efficient network management for organizations across various industries.

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